



Raptr Hacked

Raptr – the gaming community service app – has been hacked and the attackers seem to have stolen user details. <http://dgit.in/RaptrH>



KINGPIN edition

EVGA unveiled their ultimate GTX 980 Classified KINGPIN edition GPU designed for extreme overclockers.

Space Age

Despite the nuclear option being available, the researchers and scientists decided to go with the solar energy option. The primary reason from NASA was that solar panels, despite their difficulties were the more affordable option for the mission. The solar panels that'll be used in this mission have been tested over the last two years to establish them as being suitable for the environment. This strategy has already been safely tested for NASA's Juno spacecraft mission which was launched in 2011 for Jupiter on a different mission.

The current plan for the Europa Clipper calls for a 2025 launch aboard an Atlas V 551 rocket to escape Earth's gravity. There are two key trajectory options towards Jupiter – either by using the gravity from Venus on the Venus-Earth-Earth or VEEGA trajectory, which would take six years transit time, or by launching it aboard a Space Launch Vehicle which would take only three years. The Space Launch Vehicle is an in-development heavy launch system designed to fill the void left by the phasing out of the Space Shuttle. As the Clipper approaches the planet, it would then use the gravity of neighbouring moons Ganymede and Callisto to fix on a resonant orbit with Europa.

Mission and Experiments

Over the course of its 3.5 year mission (after reaching its destination) the Europa Clipper is expected to make up to 45 flybys which would bring it as close as 25 kilometers of the moon's surface. The equipment



The Space Launch Vehicle is still being developed and may be first used on the Europa Clipper mission.



At its farthest point in orbit around Jupiter, Europa is suspected to exhibit bursts of water reaching 20 times the height of Mt. Everest.

is designed to withstand up to 7 years of radiation effects before being rendered unusable, which allows for a safe 3.5 year margin within which it can fulfill its mission. A possible addition to the Clipper is that of nanoprobes which would serve a critical tertiary purpose. Since Europa is suspected to have periodic plumes of water bursting out and reaching heights of 200 kilometers these nanoprobes would measure the contents of these bursts for any indicators of life or organics.

As of now various possibilities exist as to what tests can be conducted. But the key experiments on the plans includes: shortwave infrared spectrometry to identify surface materials, ice penetrating sonography to measure the thickness of ice, stereo topographical scanning to map the moon's surface, neutral mass spectrometry to study chemical structures of the surface and atmosphere, magnetometry to characterise the moon's gravity and magnetic field, a Langmuir probe study to measure the moon's plasma field, reconnaissance photography and thermal imaging. These experiments would open up a whole new chapter in our understanding of the Solar System.

Life, the Universe and Everything

The current era of space travel began with Yuri Gagarin's debut as the first human in space. The brave Soviet cosmonaut was the first human to orbit Earth in 1961 and

since then, that is all humans have done. If the Earth was scaled down to the size of a cricket ball then humans have only ever gone a total of two meters max from the surface – to the moon. We've only travelled to the moon six times while most of our space oriented activities take place about three millimeters from the surface of that cricket ball.

In this vast, foreboding emptiness that we're a part of, we have barely begun to illuminate the darkness. It's high time we started dreaming a bit bigger. And while the Europa Clipper mission is just a prelude, it should inspire us to take aim towards greater, similar goals. Everything we know about Europa has been based on the data sets acquired from the early probes launched towards Jupiter. Now with the Europa Clipper mission finally on the horizon, scientists are excitedly looking to dig deeper into the planet's mysteries. Life as we know it is out there somewhere and



Life on Europa could be radically different, even on a micro-cellular level.

in the cricket pitch of the universe we have barely stepped on to the field.

The search for life, even if it is micro-organic or single-celled, would be a giant leap towards furthering our understanding of life, the universe and everything. It's a momentous step towards pushing our scientific, technological and imaginative abilities to the limit and to truly discover if we are alone in the universe. So, as we imagine future Europa missions, it's not entirely surprising that once we land on that surface and cut a hole through the ice, looking into the dark waters we may just find something coming up to greet us on our ice fishing expedition.